Week 3 · Laboratory Problems — build the heading autopilot in Simulink

- Course: Sensor Signal Processing and Fusion · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 3 lecture hour
- Three problems, in order. Each one adds one branch to the previous answer.

What this hour is for

Week 2 closed a loop around a **velocity** and found that the error never reached zero. This hour closes a loop around an **angle** and finds the opposite. The whole point is that the difference is **structural** — a property of the axis being controlled — and not a better controller.

The hull and the control allocation are provided. The allocation is Appendix A1's subject; Week 3 is about the loop that decides what yaw moment to demand.

Before starting

```
cd lectures/W03_simulink/problems
W03_P1_start                % creates W03_P1.slx — hull and allocation only
W03_check(1)                % run this whenever, as often as needed
```

The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it.**

> **★ One requirement on every model**
>
> A To Workspace block named `xlog`, format `Structure With Time`, fed by the plant's twelve-state output. The heading is state **12** and the yaw rate is state **6**.

Problem 1 · Proportional only (20 minutes)

Build. Command, error, one gain, into the allocation.

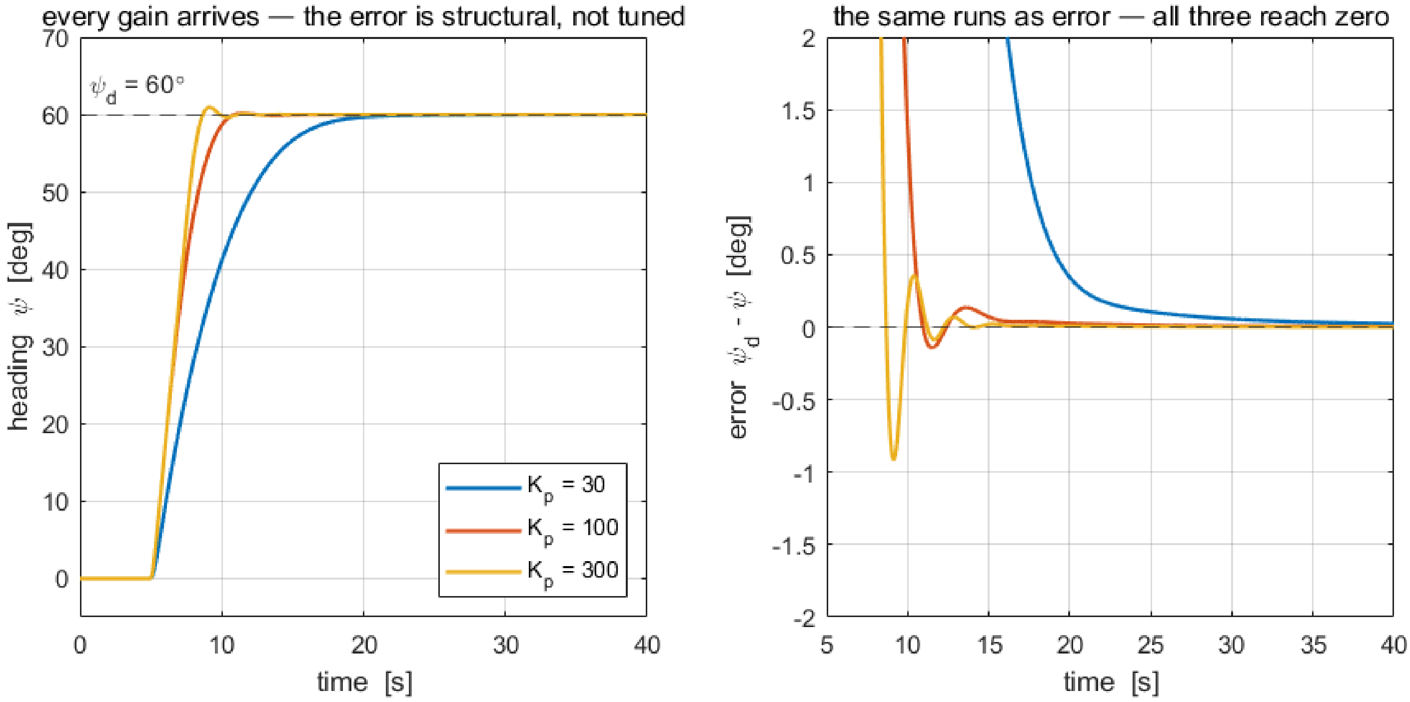
```
Step ψ_d [deg] → deg2rad → (+)(−) → Kp → τ_N → Control allocation → Otter USV
                             ↑                                          ↓
                             └──────────────  ψ  ←──────────────────────┘
```

Predict before running. Week 2's plant had no free integrator and its steady error was $u_d\,/(1+K_pK_u)$ — 44 % at $K_p = 100$. Ask the same question here **before** running: what is the steady heading error at $K_p = 30$?

Verify. `W03_check(1)`, step to 60°.

K_p	steady error	overshoot
30	0	-0.01 %
100	0	0.24 %
300	0	1.53 %

What a correct model produces



Reading the figure	
left	all three gains arrive at 60°. Higher gain arrives faster and rings more
right	the same runs as error. **All three go to zero**, and nothing was tuned to make that happen
the check	if any trace settles short of 60°, the feedback is not the heading — check that the Selector picks state 12

The point. $\psi = \int r$, so the plant carries a **free integrator** and the loop is **type 1**. Week 2's was type 0. That one structural fact — not a better controller — is why the error is zero here at every gain.

Problem 2 · Derivative action (20 minutes)

Build. One more branch: $-K_d\, r$, added to the proportional term.

> ⚠ **Feed back the yaw rate, not the derivative of the error**
>
> The two agree while ψ_d is constant and disagree at **every step**, where $\mathrm{d}\psi_d/\mathrm{d}t$ is an impulse. Differentiating the error puts that impulse straight into the actuator. The plant already **measures** r — it is state 6 — so nothing in a correct model is differentiated anywhere.

Predict before running. Substitute the law into the yaw equation:

$$M_{66}\,\ddot{\psi} + (|N_r| + K_d)\dot{\psi} + K_p\,\psi = K_p\,\psi_d$$

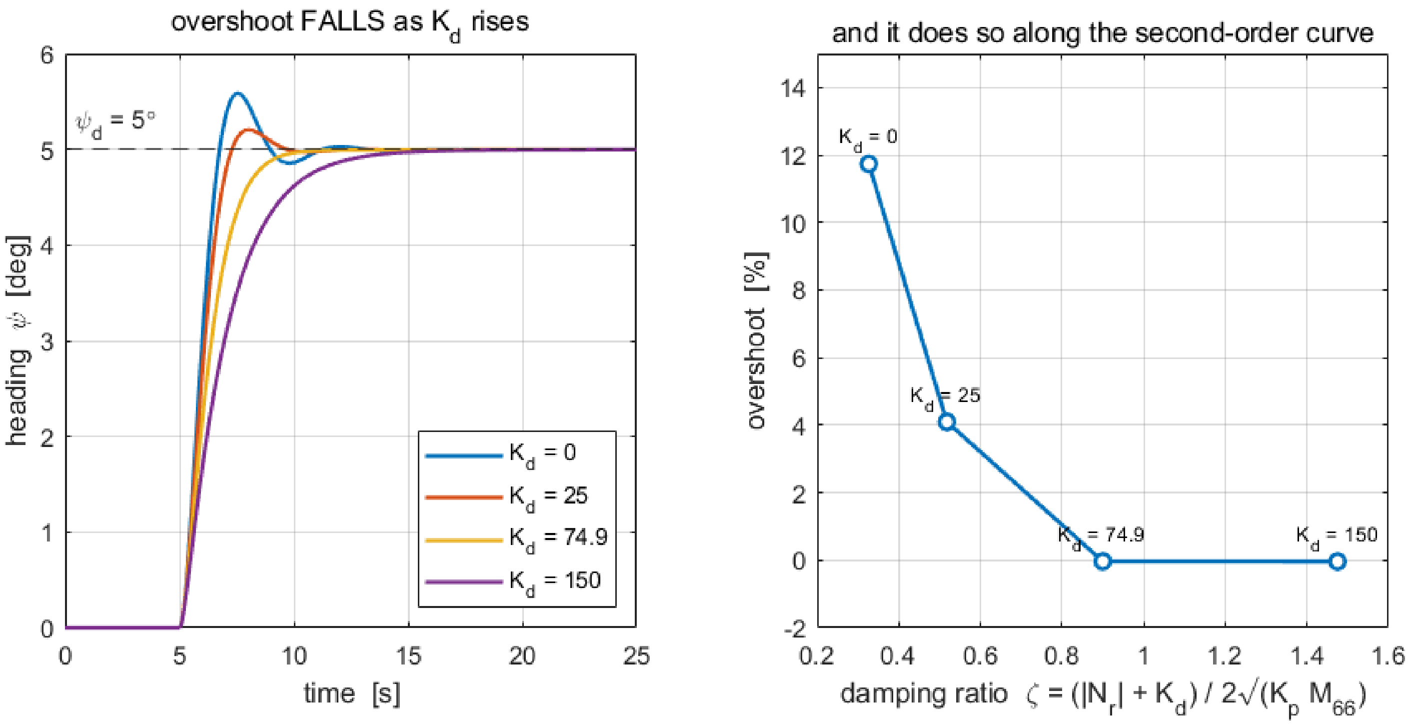
$$\omega_n = \sqrt{\frac{K_p}{M_{66}}}, \qquad \zeta = \frac{|N_r| + K_d}{2\sqrt{K_p M_{66}}}$$

Which coefficient does K_d sit beside? Answer that, and the direction of the effect follows without simulating anything.

Verify. `W03_check(2)`, $K_p = 100$, 5° step.

K_d	ζ	overshoot
0	0.327	$11.74\ \%$
25	0.518	$4.10\ \%$
74.9	0.900	≈ 0

What a correct model produces



Reading the figure	
left	four responses to the same step. **Overshoot falls as K_d rises**
right	the same four as overshoot against ζ, landing on the second-order curve
the check	if overshoot *rises* with K_d, the derivative is being taken of the error rather than fed back as r — or its sign is wrong

The point. K_d sits beside the **damping**. In Week 2 the controlled variable was a velocity, its derivative was an acceleration, and the same term sat beside the **mass**, where it made the response worse. **The term did not change; the axis did.**

Note also that the hull alone already gives $\zeta = 0.327$ at $K_p = 100$, because N_r is large. Most of the damping in this loop is not the controller's.

Problem 3 · The wrap (20 minutes)

Build. Nothing new — one line inside the control law, and a switch to turn it off.

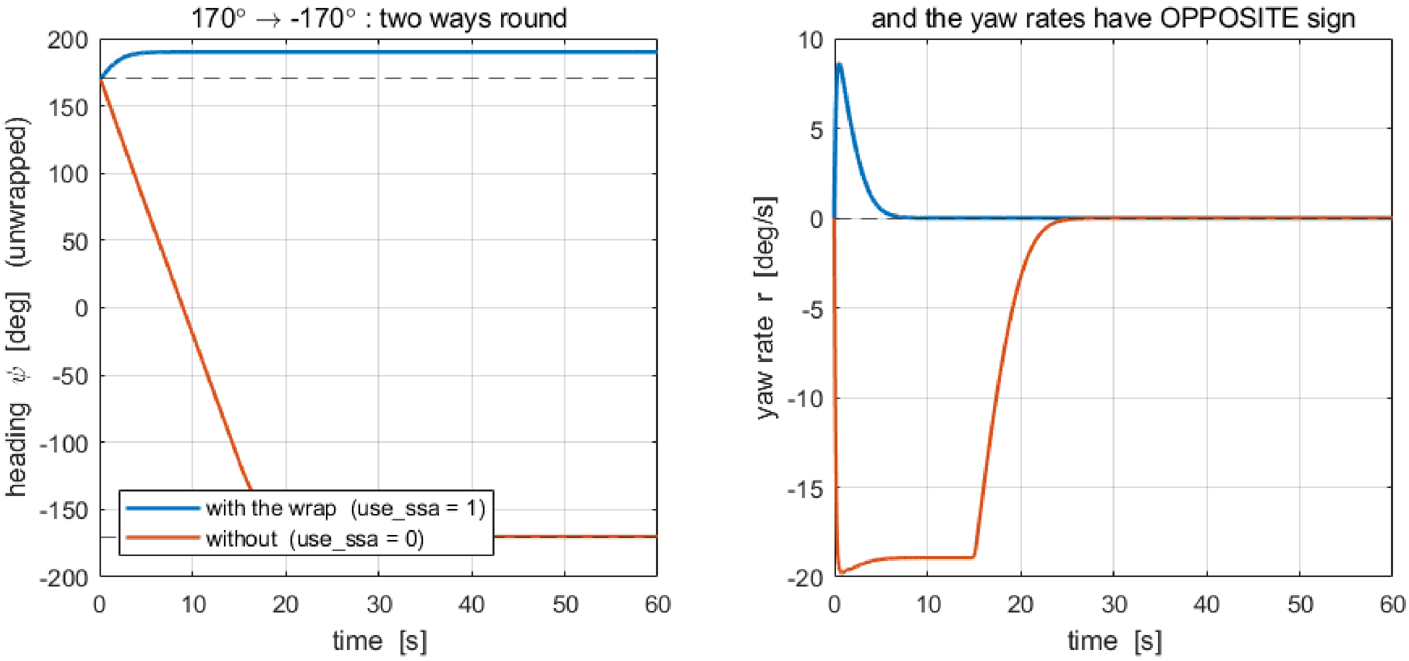
$$e = \psi_d - \psi, \qquad \mathrm{ssa}(e) = \big((e+\pi) \bmod 2\pi\big) - \pi \ \in (-\pi,\ \pi]$$

Set up the test. Start the vessel at $\psi = 170^\circ$ and command $\psi_d = -170^\circ$. The two headings are 20° **apart**.

Verify. `W03_check(3)` .

	turn executed
with the wrap	$+20^\circ$
without it	-340°

What a correct model produces



Reading the figure	
left	heading, **unwrapped**. Blue rises 20° to 190°; orange falls 340° to -170°. Both end at the same physical heading
right	the yaw rates have **opposite sign** for the whole manoeuvre
the check	if the two traces are identical, `use_ssa` is not reaching the control law

The point. Without the wrap the error is computed as -340° and the vessel goes the long way round — **seventeen times further, for the same commanded heading**. `ssa` is one line of code and it is not optional. Week 4's guidance produces commands anywhere in $(-180^\circ, 180^\circ]$, so this seam is crossed routinely.

Marking

	Weight	What is being marked
Problem 1	30	`W03_check(1)` passes; the type-1 argument is stated in one sentence
Problem 2	40	`W03_check(2)` passes; the answer to "which coefficient does K_d sit beside" is stated **and** the rate feedback is r, not $\mathrm{d}e/\mathrm{d}t$
Problem 3	30	`W03_check(3)` passes; the cost of omitting `ssa` is quantified

If something goes wrong

Symptom	Cause	Fix
`The model has no To Workspace block whose variable name is xlog`	the variable name is still `simout`	rename it
Heading settles short of the command	the feedback is not state 12	the Selector index must be 12 for ψ, 6 for r
The vessel spins continuously	the error sign is reversed	the sum is $\psi_d - \psi$, not $\psi - \psi_d$
Overshoot rises with K_d	the derivative is taken of the error	feed back r directly; the sign is $-K_d r$
Huge spike in the actuator at each step	same cause	same fix
The two Problem 3 traces are identical	`use_ssa` never reaches the law	wire it in as a Constant, like K_p and K_d
Angles look 57 times too large or small	degrees and radians mixed	every angle inside the loop is in **radians**; convert once, at the command

Reference answers are in `../solutions/`. Read them **after** attempting the problem.